Rational Functions: Final Retake Practice

Name: _____ Date: _____

13 questions | 58 points | 75 minutes

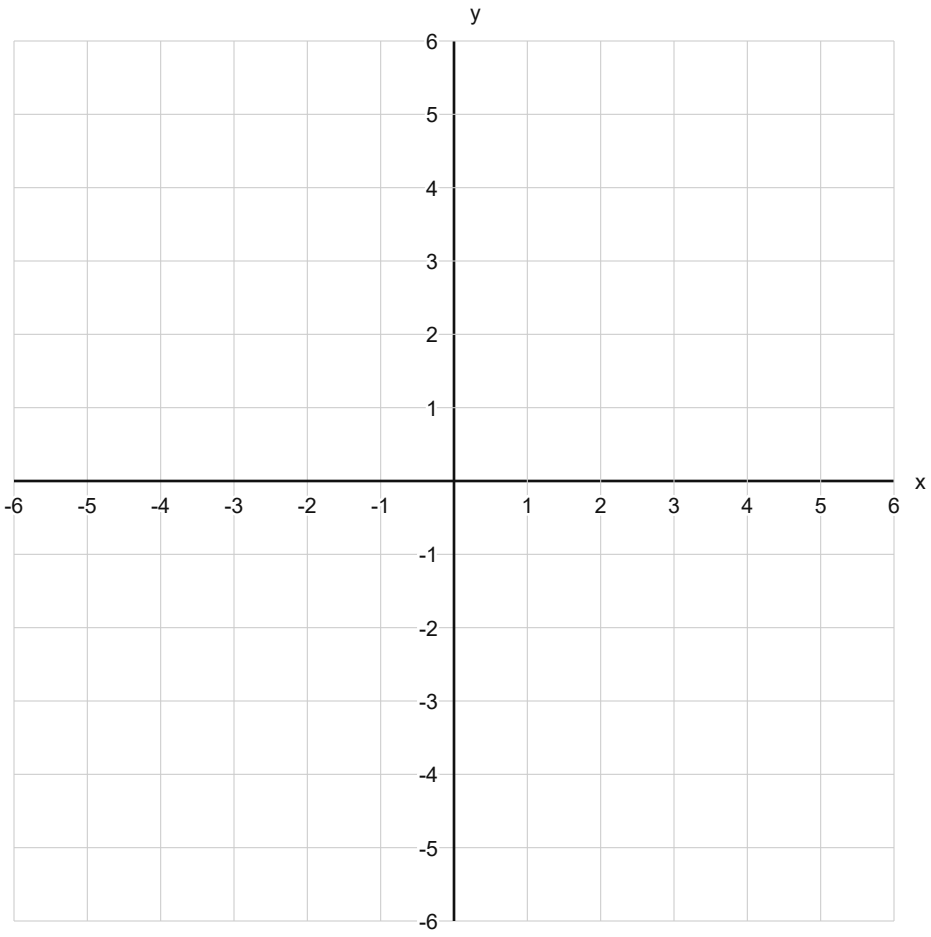
Complete every labeled part. Show the algebra and explanations requested. Give exact answers unless an estimate is requested. A four-function calculator is permitted. You have 75 minutes total.

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1. 8 points

Let $f(x) = \dfrac{x+1}{2-x}$.

(a) Graph the function. Label its vertical and horizontal asymptotes and two points, one on each branch.



(b) State the far-left and far-right end behavior.

Far-left behavior	Far-right behavior

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2. 4 points

Simplify $\frac{2}{x^2+x-6} - \frac{3}{x^2+5x+6}$. State all excluded real inputs. Show your algebra.

Work and final answer

Algebra
Simplified expression
Excluded real inputs

3. 4 points

Simplify $\frac{x^2-9}{x^2+5x+6} \cdot \frac{x^2+4x+4}{x^2-6x+9}$. State all excluded real inputs. Show your algebra.

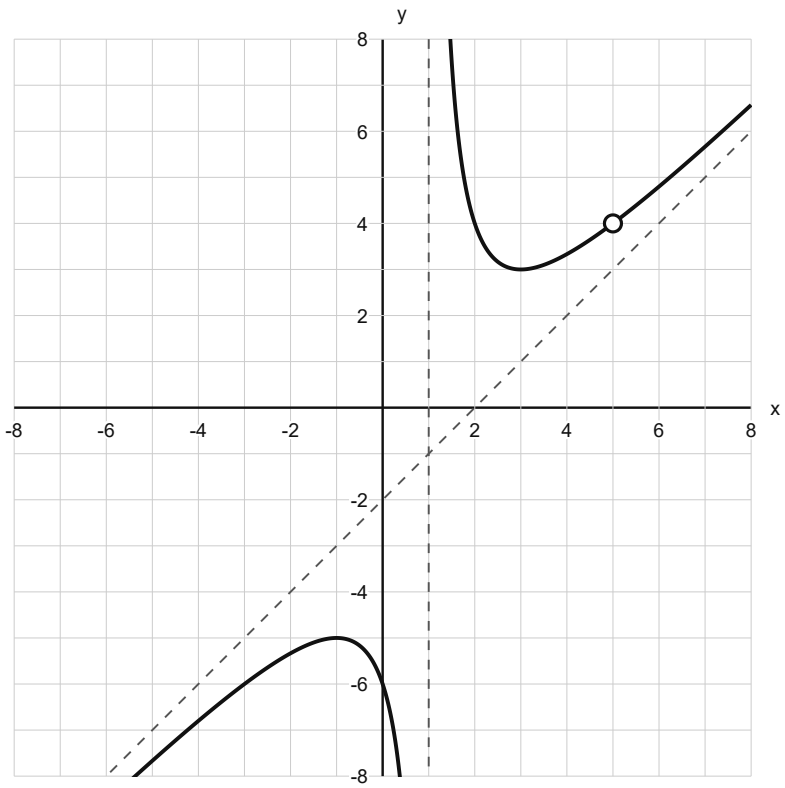
Work and final answer

Algebra
Simplified expression
Excluded real inputs

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4. 8 points

Use the supplied graph. The curve continues beyond the window; an open circle denotes a missing point. Give intervals in interval notation.



(a) Where is the function increasing, and where is it decreasing?

Increasing
Decreasing

(b) Give every relative maximum and relative minimum.

Relative maximum
Relative minimum

(c) Give the intervals where the graph is concave up and concave down.

Concave up
Concave down

(d) Give each asymptote equation, or state that none exists.

Vertical
Horizontal
Slant

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5. **10 points**

Let $R(x) = \dfrac{-3x^2 + 12x - 9}{2x^2 + 2x - 4}$.

(a) Find all intercepts, holes and asymptotes. Give exact coordinates or equations as appropriate. Show your algebra and state the original domain exclusions.

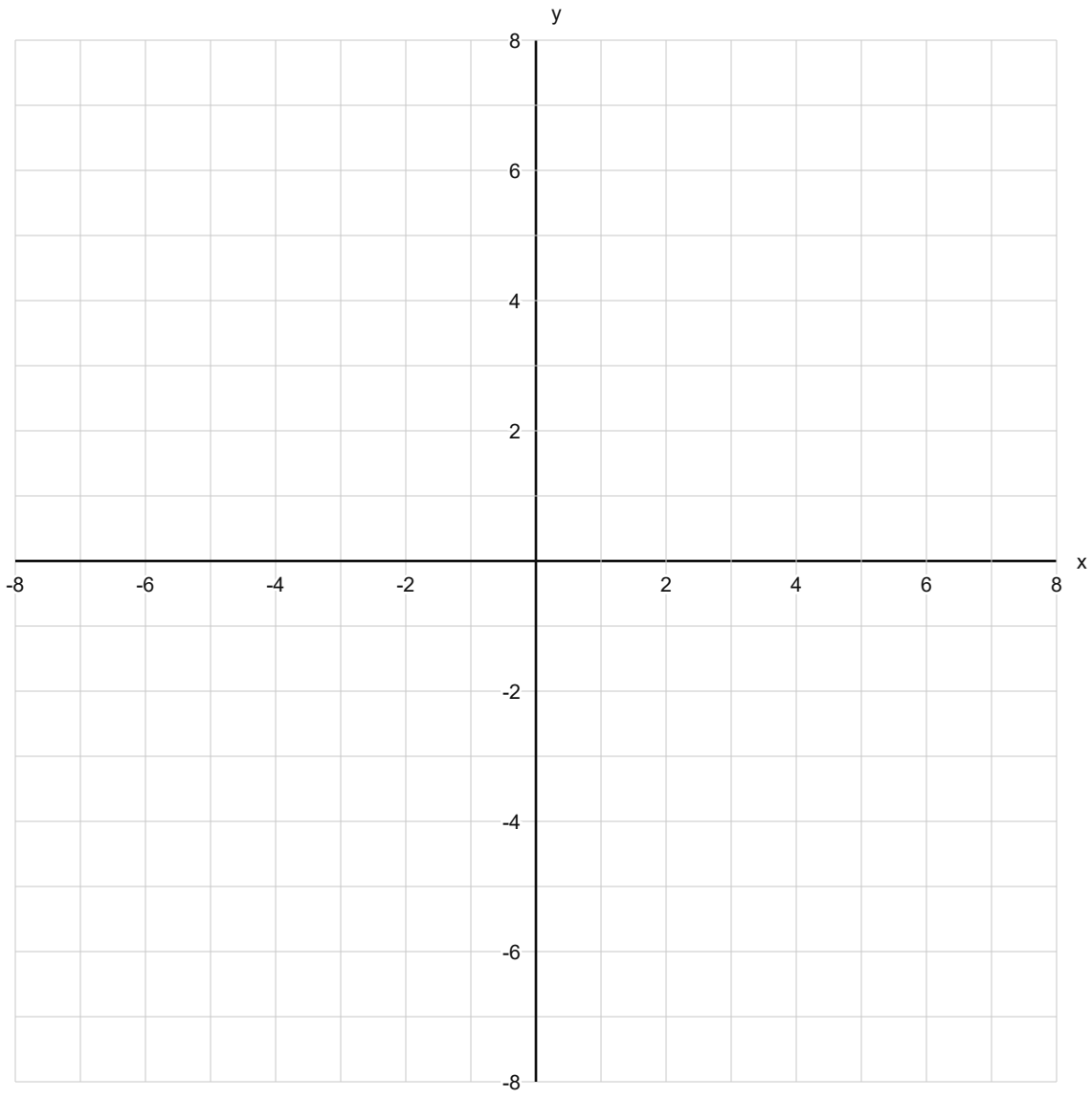
Algebra and original exclusions
x-intercept(s)
y-intercept
Hole(s)
Vertical asymptote(s)
Horizontal or slant asymptote(s)

Continue with part (b) on the next page.

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5. (continued)

(b) Sketch the function on the grid and label all features found in part (a).



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6. 4 points

Let $S(x) = \dfrac{x^2 + 2x - 7}{2x - 2}$.

(a) Find each horizontal or slant asymptote, or state that none exists. Justify algebraically.

Algebra	
Horizontal asymptote(s)	Slant asymptote(s)

(b) As x tends to negative infinity and as x tends to positive infinity, state what S(x) tends to. Explain algebraically why the curve approaches its asymptote.

Far-left behavior	Far-right behavior
Explanation	

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7. 4 points

A container initially holds 5 liters containing 10 grams of dissolved salt. A solution flows in at 2 liters per minute, adding 6 grams of salt per minute; nothing flows out. After $t \geq 0$ minutes, the concentration is $C(t) = \dfrac{10 + 6t}{5 + 2t}$ grams per liter. Interpret the long-term concentration. State whether it approaches that value from above or below and whether it reaches it at any finite time. Justify your conclusions.

Answer and justification

Long-term value, units and meaning
Approach from above or below
Finite attainment and justification

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8. 4 points

Construct a rational function whose only x-intercept is $(-2, 0)$, whose only vertical asymptote is $x = 3$, whose horizontal asymptote is $y = -2$, and whose only hole is at $x = 0$.

(a) Write your function in factored form and verify each required feature, including the original domain exclusions.

Function
Verification of features and exclusions

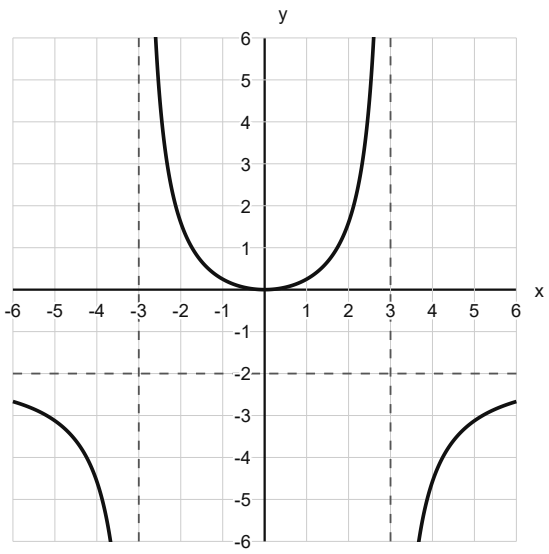
(b) Give the hole coordinates for your chosen function. Show your calculation.

(c) Does your function have a y-intercept? Explain.

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9. 2 points

The graph represents a nonzero rational function. Which degree relationship must hold? Select one answer and justify it using the graph.



A. The denominator degree is greater than the numerator degree.

B. The numerator degree is exactly one greater than the denominator degree.

C. The numerator and denominator degrees are equal.

D. The numerator degree is exactly two greater than the denominator degree.

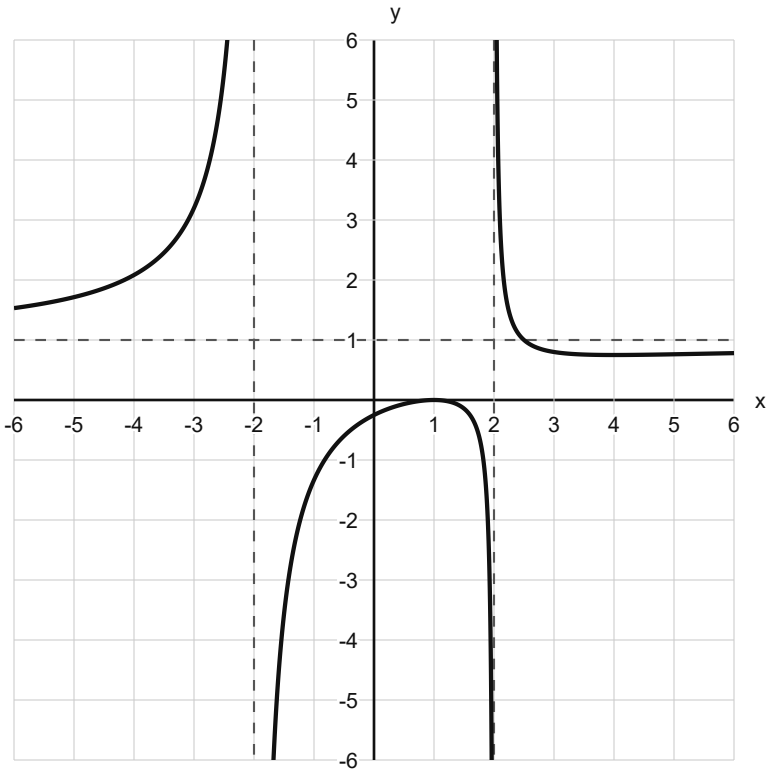
Answer

Choice	Justification

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10. 2 points

Which equation matches the entire graph? Select one answer. Justify using the behavior at the x-intercept and another graph feature that distinguishes your choice from the remaining look-alike choices.



A. $y = \dfrac{x-1}{x^2-4}$

B. $y = \dfrac{-(x-1)^2}{x^2-4}$

C. $y = \dfrac{(x-1)^2}{2x^2+8}$

D. $y = \dfrac{(x-1)^2}{x^2-4}$

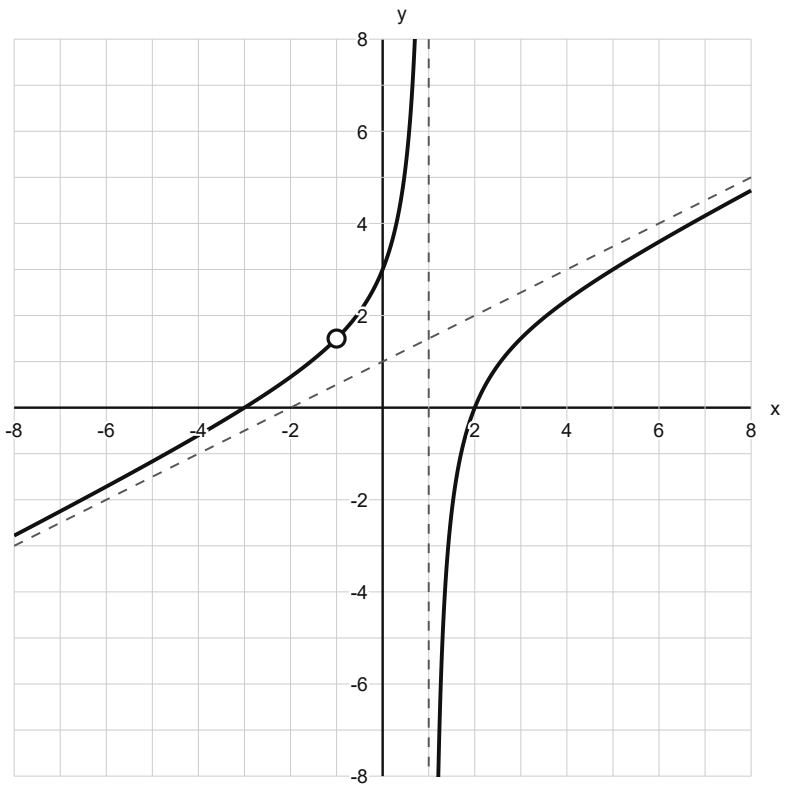
Answer

Choice	Justification

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11. 2 points

Which function matches the entire graph? The curve continues beyond the window. Select one answer. Include the missing point's coordinates and one other distinguishing feature in your justification.



A. $y = \dfrac{(x-2)(x+3)}{2(x-1)}$

B. $y = \dfrac{(x+1)(x-2)(x+3)}{2(x+1)(x-1)}$

C. $y = \dfrac{(x+1)(x-2)(x+3)}{2(x+1)^2}$

D. $y = \dfrac{-(x+1)(x-2)(x+3)}{2(x+1)(x-1)}$

Answer

Choice	Justification

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12. 2 points

Let $f(x) = 3x^4 - x + 2$ and $g(x) = \dfrac{4x^m - 7x^6 + 3}{x^2 + 1}$, where m is a positive integer. Which value makes the far-left and far-right directions of g match those of f? Select one answer and justify it.

A. $m = 8$

B. $m = 7$

C. $m = 6$

D. $m = 2$

Answer

Choice	Justification

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13. 4 points

A rational function H has only two undefined real inputs, -2 and 1. One is a hole and the other is a vertical asymptote. Selected values are shown; decimals are rounded. Identify the likely x-coordinate of each and explain the table evidence. At the vertical asymptote, include the behavior from each side.

x	-2.1	-2.01	-2.001	-2	-1.999	-1.99	-1.9
H(x)	102	10002	1000002	undefined	1000002	10002	102

x	0.9	0.99	1	1.01	1.1
H(x)	2.1189	2.1119	undefined	2.1104	2.1041

(a) Hole

Location	Evidence

(b) Vertical asymptote

Location	Evidence and both one-sided behaviors